

title: I · Privacy Impact Assessment (RA 10173) description: Draft Privacy Impact Assessment for the DepEd HRIS in line with RA 10173 (Data Privacy Act of 2012) and NPC Circular 16-01 — scope, data inventory, lawful bases, data-flow diagrams, subject rights, risk matrix, technical and organisational measures, breach protocol, retention schedule, and DPO sign-off sequence.

I · Privacy Impact Assessment (RA 10173)

!!! info “About this paper” The National Privacy Commission (NPC) requires a **Privacy Impact Assessment (PIA)** for any personal-data processing system operated by a personal-information controller (PIC). DepEd is a PIC. This paper is a **draft PIA** built from the scope in [Paper A](#), the architecture in [Paper C](#), and the migration in [Paper H](#). It is designed to be **filed with the NPC** after review by DepEd’s Data Protection Officer (DPO) and edited with organisation-specific facts (DPO name, filing date, incident-response contacts).

I.1 Regulatory anchors

The PIA structure follows:

- **RA 10173** — Data Privacy Act of 2012, especially §11 (general principles), §12 (lawful basis for processing personal information), §13 (lawful basis for sensitive personal information), §16 (rights of data subjects), §19 (data-protection officer), §20 (security measures), §38 (penalties).
- **NPC Circular 16-01** — Security of Personal Data in Government Agencies.
- **NPC Circular 16-03** — Personal Data Breach Management.
- **NPC Circular 2022-04** — Registration of Data Processing Systems.
- **NPC Advisory 2017-01** — Designation of Data Protection Officers.
- **ISO/IEC 27001** and **ISO/IEC 27701** control mappings as adopted in Paper C §7.

I.2 Scope of the PIA

In scope. All processing performed by the DepEd HRIS as scoped in [Paper A](#) — the 11 functional modules, the applicant portal, offline mobile / kiosk access, SMS / USSD channels (D §1), the HR Copilot (D §3), the payroll anomaly detector (D §4), the hash-chained audit ledger (D §5), the transparency portal (D §6), and all nine regulator integrations. **Data migration from legacy PIS** ([Paper H](#)) is explicitly in scope.

Out of scope. DepEd systems outside the HRIS (LIS, EBEIS, DepEd Commons), personal devices of employees not enrolled as endpoints, non-personal statistical outputs.

Filing intent. File once with the NPC upon go-live (M7), then re-file **within 30 days** of any material change to processing (per NPC Circular 2022-04).

I.3 Roles

Role	Assignment	Responsibility
Personal Information Controller (PIC)	Department of Education	Purposes and means of processing
Data Protection Officer (DPO)	To be designated by DepEd Secretary	RA 10173 §19 responsibilities; sign-off on this PIA
Compliance Officer for Privacy (COP)	Per RO / SDO	Local privacy compliance
Personal Information Processor (PIP)	Prime contractor (bidder) + hosting subcontractors (none per SCC 7)	Processing under DPA
NPC	National Privacy Commission	Oversight; PIA registration

Since SCC clause 7 forbids subcontracting, the PIP layer collapses to the prime contractor plus DepEd's own operations team.

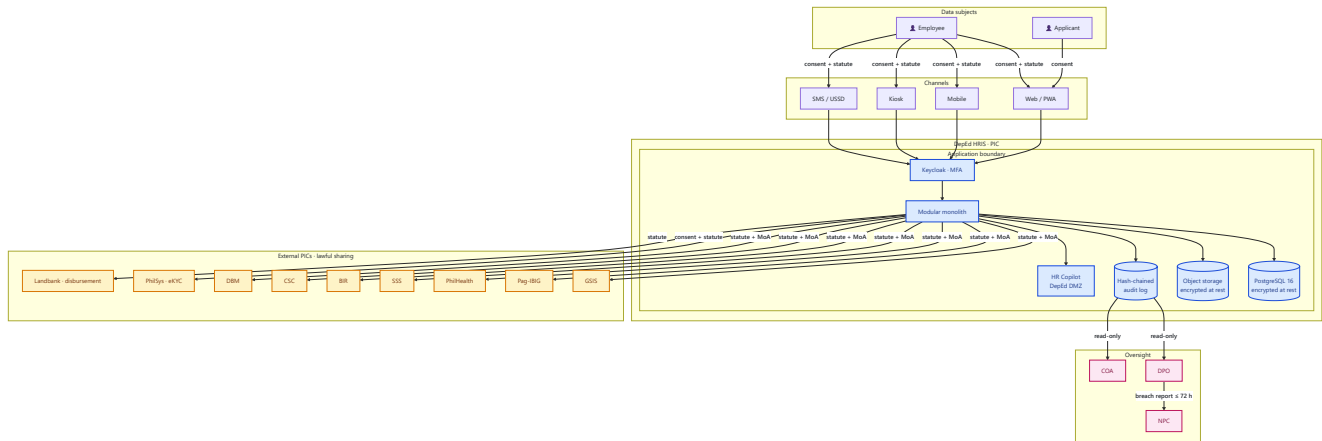
I.4 Data inventory — categories and sensitivity

Personal data processed, classified per RA 10173 §3 (definitions):

Category	RA 10173 class	Subjects	Volume	Purpose
Employee identity — name, PhilSys number, birthdate, birthplace, gender	Personal (§3.g)	Employees + applicants	~ 1.1 M	Identity resolution, authentication, appointment
Contact — address, phone, email, emergency contact	Personal	Employees + applicants	~ 1.1 M	Notifications, HR contact
Government IDs — GSIS, Pag-IBIG, PhilHealth, SSS, BIR TIN	Personal + Sensitive (regulatory implication)	Employees	~ 1.0 M	Regulator remittance
Employment history — appointments, plantilla items, assignments, promotions	Personal	Employees	~ 1.0 M	Personnel management
Compensation — salary, allowances, deductions, deductions history	Personal	Employees	~ 1.0 M	Payroll
Bank account — account number, bank code	Personal + Sensitive	Employees	~ 900 K	Disbursement
Performance evaluations — SPMS, IPCR, OPCR, disciplinary records	Sensitive (§3.I.4) — social security / due-process	Employees	~ 1.0 M	Performance mgmt
Health profile — HMO, medical clearance, wellness	Sensitive (§3.I.1) — health	Employees	~ 1.0 M	Wellness module (§5.10)
Educational credentials — degrees, certifications	Personal	Employees + applicants	~ 1.5 M	Eligibility
Eligibility & training — CSC eligibility, training records	Personal	Employees + applicants	~ 1.5 M	Career progression
Biometric data — fingerprint / face templates for time & attendance	Sensitive (§3.I.3) — biometrics	Employees	~ 900 K	Attendance
Applicant PDS + attachments	Personal + potentially Sensitive	Applicants	~ 200 K/yr	Recruitment (§5.1)
Complaints / disciplinary — CSC cases, admin cases	Sensitive (§3.I.2) — proceedings	Small subset	~ variable	Personnel management
System usage — audit log, access log	Personal (§3.g if traceable)	All users	very large	Audit, security
HR Copilot conversations (D §3)	Personal (varies by query)	Users of Copilot	~ variable	Assistance

Sensitive personal information (RA 10173 §13) is processed on multiple bases — see §1.6.

I.5 Data-flow diagram



Key facts encoded in the diagram:

- Employees and applicants are the only data subjects (RA 10173 §3.c).
- All ingress traverses Keycloak with MFA (RA 10173 §20 · organisational).
- Data at rest is encrypted (RA 10173 §20 · technical).
- The HR Copilot runs **inside the DepEd DMZ** — no data leaves the perimeter (D §3).
- Every regulator share has a **defined lawful basis** — either statute or MoA (see §I.6).
- The audit log is **hash-chained** (D §5), giving the DPO and COA tamper-evident visibility.

I.6 Lawful basis matrix — per processing purpose

Per RA 10173 §12 (personal) and §13 (sensitive), every processing activity is mapped to at least one lawful basis:

Processing purpose	Data used	Personal · §12 basis	Sensitive · §13 basis
Personnel management (§5.1, 5.6)	Identity, employment, compensation	§12(a) consent · §12(e) public function	§13(b) legal obligation (Civil Service laws)
Payroll disbursement	Identity, compensation, bank	§12(b) contract of employment · §12(e)	§13(b) legal obligation (GAA, DBM)
Regulator remittance (GSIS / Pag-IBIG / PhilHealth / SSS / BIR)	Government IDs, compensation	§12© legal obligation	§13(b) legal obligation
Recruitment (§5.1–5.3)	PDS, applicant data	§12(a) consent	§13(a) explicit consent
Time & attendance	Biometric templates	§12(e) public function	§13(b) legal obligation + §13(a) consent
Performance evaluations	SPMS, IPCR, OPCR	§12© legal obligation	§13(b) legal obligation (CSC rules)
Wellness & health (§5.10)	Health profile	§12(a) consent	§13(a) explicit consent
Disciplinary cases	Complaint records	§12© legal obligation	§13(f) protection of lawful rights
PhilSys eKYC (D §13)	PhilSys number + demographic verify	§12(a) consent · §12©	§13(a) explicit consent
Audit & security	Access logs, audit trail	§12(f) legitimate interest	n/a
HR Copilot Q&A (D §3)	Query text + retrieval context	§12(a) consent	§13(a) explicit consent
Transparency portal (D §6)	Anonymised appointment feed	§12(f) legitimate interest + PIC-controlled anonymisation	n/a — data anonymised before publication
Legacy migration (Paper H)	Everything in legacy PIS	§12© legal obligation (continuity of public function)	§13(b) legal obligation

Every processing activity in the running system carries a link back to a row in this matrix, exposed to the DPO through the audit ledger.

I.7 Data-subject rights (RA 10173 §16) — implementation

Each right in §16 must be honoured. Implementation surface:

Right	Where implemented	How
§16(a) To be informed	ESS onboarding + privacy notice on every collection form	Explicit “you are giving us X for purpose Y under basis Z” text
§16(b) To object	ESS settings page	Employee can flag processing they object to; escalates to DPO
§16(c) Access	ESS “My 201 File” page	Complete personal-data export in machine-readable form
§16(d) Correct	ESS “Update my info” form	Change request routes to HR officer for verification
§16(e) Erase / block	Contact DPO channel	Handled by DPO under §16(e) exceptions (e.g. legal-obligation retention)
§16(f) Damages	DPO channel + NPC complaint	Documented process, escalation SLA
§16(g) Data portability	ESS “Export my data” (JSON + PDF)	Structured export under NPC Circular 2016-02
§16(h) Right to complain to NPC	Prominently displayed on every screen footer	Direct link to NPC complaint form

I.8 Risk assessment — likelihood × impact

Per NPC guidance, risks are scored on 1–5 scales (5 = highest). Only risks ≥ 12 receive dedicated mitigation.

#	Risk	Likelihood	Impact	Score	Primary control
1	Payroll disbursement to wrong account	3	5	15	M6 parallel-run + anomaly detector (D §4)
2	Unauthorised regulator remittance	2	5	10	Signed API calls, per-regulator audit log
3	Insider access abuse	3	4	12	RBAC + ABAC + RLS + audit ledger + quarterly access review
4	External breach via app vulnerability	2	5	10	Pen-test at M5 + quarterly, WAF, CSP, dependency scanning
5	Sensitive-data leak via HR Copilot	2	4	8	Self-hosted model, per-role RAG scoping, red-team eval (D §3)
6	Migration-time data loss / corruption	3	5	15	Paper H waves + reconciliation gates + rollback
7	Lost or stolen mobile / kiosk device	4	3	12	Device attestation + short session TTL + remote wipe
8	Session hijack	2	4	8	MFA + short-lived tokens + IP anomaly detection
9	Backup exposed on wrong tier	2	5	10	Encryption-at-rest, access reviews, offline copies
10	Print / screenshot exfiltration	4	3	12	Watermarking, download audit, disciplinary policy
11	Regulator forced disclosure	2	3	6	Legal review before disclosure; DPO in the loop
12	Third-party dependency compromise	2	4	8	SBOM, dependency signing, minimal supply-chain surface
13	Employee-signed PhilSys eKYC replay	2	4	8	Nonce-bound requests, PhilSys audit logs
14	Transparency-portal re-identification	3	4	12	k-anonymity + differential privacy for public feeds (D §6)
15	Retention exceeded on separated employees	3	3	9	Automated retention policy (§I.11)

I.9 Technical and organisational measures (RA 10173 §20)

Consolidated from the design in [Paper C §7](#), the deployment view in Paper C §8, the migration plan in Paper H, and the runbooks in the operations manual (planned).

Technical (§20.c)

- **Encryption in transit** — TLS 1.3 minimum; internal traffic mTLS through service mesh.
- **Encryption at rest** — PostgreSQL TDE (or LUKS + column-level for sensitive), object storage server-side encryption, backups encrypted with a rotated KMS key.
- **Authentication** — Keycloak OAuth2 / OIDC, MFA mandatory for HR / Payroll / Admin roles, single sign-out.

- **Authorisation** — RBAC × ABAC, row-level security per organisational scope, per-request policy check.
- **Audit** — every mutation captured to hash-chained ledger (D §5) with a per-year Merkle-root anchor.
- **Segmentation** — application, database, and admin planes on separate VLANs; production isolated from non-prod.
- **Deployment residency** — [Paper F](#) prices **two deployment options** (on-premises in DepEd facilities, or public cloud on GovCloud PH / AWS Manila). **Both satisfy RA 10173 residency** when configured per §I.5. Option B (cloud) requires explicit region pinning to a PH-based region and an annual review of the provider's SOC 2 / ISO 27001 attestation. Option A (on-prem) satisfies residency by construction — data never leaves DepEd-owned premises.
- **Backup & recovery** — 3-2-1 rule, tested restore quarterly, RTO 4 h / RPO 15 min.
- **Endpoint hardening** — CIS benchmarks on all servers, EDR on admin workstations.
- **Vulnerability management** — SCA and SAST in CI, DAST pre-release, third-party pen-test twice per year.
- **Anomaly detection** — payroll (D §4), authentication (rate + IP), data exfiltration.
- **Data minimisation** — API responses trimmed per role; download watermarking; report-side redaction.

Organisational (§20.a, §20.b)

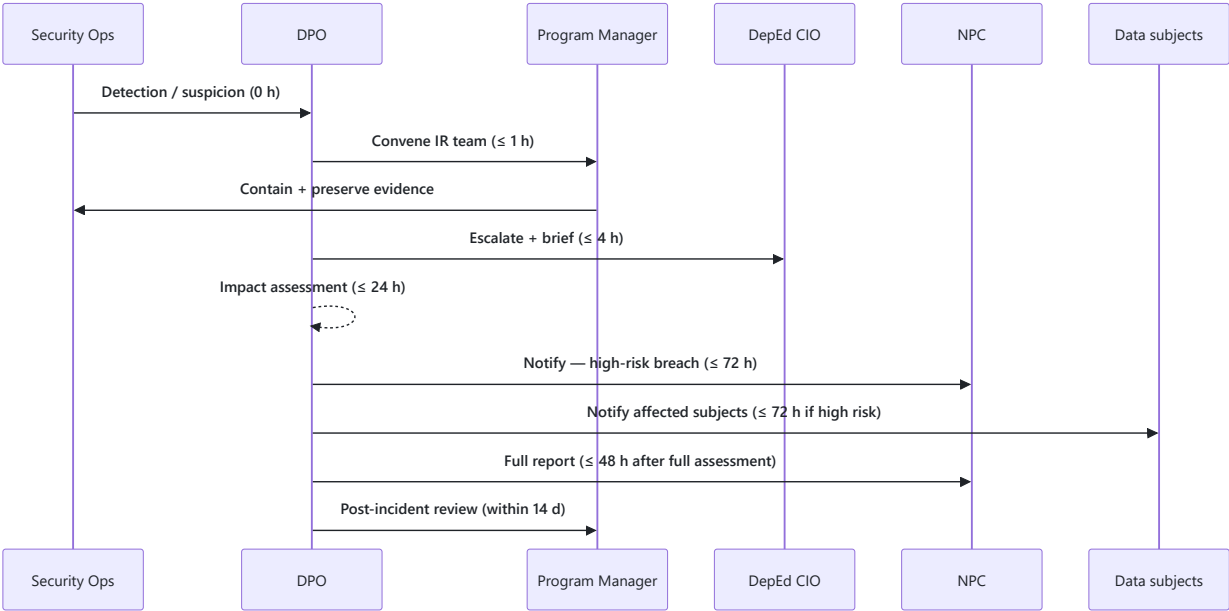
- **Designated DPO** with direct line to DepEd Secretary; contact published on every user-facing screen.
- **Per-RO / SDO Compliance Officers for Privacy** (COPs), trained on RA 10173 and NPC circulars.
- **Data-processing register** maintained per NPC Circular 2022-04.
- **Vendor management** — no subcontracting (SCC 7); prime contractor's team subject to background checks.
- **Training** — mandatory RA 10173 module for every employee with system access, annual refresher.
- **Access reviews** — quarterly HR + IT joint review of privileged accounts.
- **Incident-response plan** — see §I.10.
- **PIA refresh cadence** — annually, and on any material change to processing.

Physical (§20.a)

- **Data-centre access** — biometric + card + escort for non-staff.
- **CCTV** at all racks and access doors.
- **Environmental** — fire suppression, redundant power, cooling.
- **Media disposal** — degaussing + certified destruction with COA-visible chain of custody.

I.10 Breach management — RA 10173 §38 + NPC Circular 16-03

Trigger: any suspected unauthorised access to, disclosure of, or loss of personal data.



Templates for the breach notification (both NPC and subject-facing) are held in the DPO’s runbook and drilled twice per year.

I.11 Retention & disposal schedule

Per COA and CSC retention rules, with NPC-guided minimisation. Retention starts on the trigger event.

Data category	Trigger	Retention
Employee 201 file	Separation	10 years then archived
Payroll records	Fiscal year close	10 years then archived
Applicant PDS (not hired)	Application closed	2 years then delete
Applicant PDS (hired)	Hire date	Merged into 201 file (retention as above)
Biometric templates	Separation	1 year then delete (template only; audit log retained)
Audit log	Log creation	7 years hot; longer archived per DepEd + COA
Session logs	Session end	90 days then delete
HR Copilot conversations	Conversation end	30 days then delete unless flagged for review
Backup snapshots	Snapshot creation	90 days rolling + monthly for 12 months + yearly for 7 years
Migration audit log	Cutover completion	10 years cold storage

I.12 DPO sign-off sequence

This PIA is not final until:

- 1. **DPO review** — DPO reviews §§I.1–I.11, edits any organisation-specific facts.
- 2. **DepEd CIO review** — architecture and controls check against the operating context.
- 3. **Legal review** — DepEd Legal Service reviews lawful bases and retention schedule.
- 4. **Program Manager sign-off** — confirms deliverability inside the 365-day contract.
- 5. **NPC filing** — per NPC Circular 2022-04, filed within 30 days of go-live.

6. **Annual re-affirmation** — DPO reviews and re-affirms every year, and on any material change.

I.13 What is deliberately not covered here

- **PIAs for other DepEd systems** (LIS, EBEIS, DepEd Commons) — separate PIAs, out of scope.
- **Employee-personal-device policies** — separate HR issuance, references RA 10173 but is not this PIA.
- **Third-party analytics** — none used; explicitly excluded from architecture.
- **International data transfer** — none by design (HR Copilot, cloud DR, all remain in-country); if this ever changes, this PIA must be re-filed with §§I.4–I.6 updated for the transfer.

I.14 Cross-references

- Scope of processing → [Paper A · Brief](#)
 - Technical controls → [Paper C · Architecture §7 Security & Privacy](#)
 - Migration processing → [Paper H · Data Migration](#)
 - Team accountable for security & DPO liaison → [Paper F §F.3](#)
 - Risk register anchor → [Paper B §15](#) (R6, R10)
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This is a draft PIA prepared as an independent analysis. Before filing with the NPC, the designated DPO must complete §§I.3 (their name and contact), I.9 (organisation-specific procedures), I.10 (contact details for the IR team), and I.12 (sign-off dates).